

1. Pre-requisites for Full Closed Loop

V 3.4

Please note that with autoISF you are in an early-dev. environment, where the user interface is **not optimized for safety** of users who stray away from intended ways to use. Good safety features exist, but these are only as good as the development-oriented user understands and implements them. This is not a medical product, refer to disclaimer in [section 0](#)



- 1.1 Well tuned hybrid closed loop
- 1.2 Fast insulin
- 1.3 Reliable insulin delivery from pump and cannula
- 1.4 Excellent CGM
- 1.5 Meal-related limitations?
- 1.6 Lifestyle-related limitations?
- 1.7 Time required for setting-up

Available related case studies:

- Case study 1.1: Occlusion
- Case study 1.2: Comparing insulins for FCL
- Case study 1.3: Jumpy CGM
- Case study 1.4: Lost pump connection
- Case study 1.5: Overlapping 2 x G6
- Case study 1.6: When catastrophe strikes
- Case study 1.7: [Libre3 \(1 minute\)](#) placeholder

1.1 Well-tuned hybrid closed loop (HCL)

It is advisable to first establish a well-tuned **hybrid closed loop** before considering the transition to FCL. Best if you achieved good HCL performance **without** using Autotune or dynamicISF (which can introduce, or cover up, problems that would get exposed in your transition to Full Closed Loop (FCL); more see at beginning of [section 4](#)).

There are three important reasons for starting on a solid basis (profile):

- The UAM full closed loop requires a highly personalized (individual) tuning of settings, so the loop will give insulin mimicking YOUR successful hybrid closed loop mode.
- Key autoISF tuning parameters multiply with your profile_ISF. You could be “big time starting on the wrong foot”, and potentially waste a lot of tuning time with your autoISF! See for example: red box in [section 4.3.2](#)
- The UAM full closed loop comes with new parameters to be set and tuned. It would be problematic to set and tune several new parameters before the basics were tuned “right”. Errors could easily be balanced with counter-errors. This can work in single scenarios, but would create a highly unstable system. It would be very hard to re-calibrate better later, too.

1.2 Fast insulin (Lyumjev, Fiasp, Apidra?)

It should be clear without saying, but it is absolutely necessary to feed your loop with correct time-to-peak and DIA for your insulin, so it has any chance to know how the job will get active in your body. See “Insulin_DIA...pdf” in: <https://github.com/bernie4375/HCL-Meal-Mgt.-ISF-and-IC-settings>

39 If the user does not bolus for meals, clearly a very fast insulin is needed so, upon realization of a
40 starting meal-related glucose rise, the loop has any chance to eventually keep glucose in range (by
41 common definition, under 180 mg/dl (10 mmol/l))

42

43 A modelling study..

44 key findings are summarized in initial section of [case study 1.2](#); for more see: "The artificial
45 pancreas and meal control...pdf" in: [https://github.com/bernie4375/HCL-Meal-Mgt.-ISF-and-IC-](https://github.com/bernie4375/HCL-Meal-Mgt.-ISF-and-IC-settings/blob/HCL-.settings-main-repo-(pdf)/)
46 [settings/blob/HCL-.settings-main-repo-\(pdf\)/](https://github.com/bernie4375/HCL-Meal-Mgt.-ISF-and-IC-settings/blob/HCL-.settings-main-repo-(pdf)/)

47 ...can show in quantitative terms, that **faster insulins**:

- 48 • will result in significantly **lower** glucose **peaks** than slower insulins
- 49 • **tolerate** a couple of minutes **delayed** first meal bolus while not incurring unacceptable
50 height of peaks
- 51 • minimize the effect on glucose peak from **different** carb load (**meal sizes**).

52 In conclusion, do not attempt FCL with other insulin than Lyumjev® or Fiasp®.

53 Potential exceptions:

- 54 ○ Being consistently on low carb, on a GLP-1 drug, or with gastroparesis -all of which ease the
55 job of a FCL significantly.
- 56 ○ According to [case study 1.2](#), Apidra® might work, too, but Humalog® would not work well
57 (with "normal diet").

58

59 1.3 Reliable insulin delivery from the used pump/cannula/insulin system

60

61 **Good tolerance of Lyumjev (or Fiasp): Occlusions threaten the function of the full closed loop.**

62

63 It is very important to have an eye on the time a **cannula (or pod)** is in use (many find **48 hrs** to be
64 the **limit**), and whether hard-to-explain glucose rises happen at ever increasing „fake“ iob (even
65 before a 48 hr routine replacement). (You easily lose 25% TIR that day, see [case study 1.1](#))

66 It is absolutely contra-indicated to attempt FCL coming from leaking pods and associated erratic
67 sensitivity swings that may or may not have been somewhat controlled and tolerable by
68 dynamicISF or other measures when you were Hybrid Closed Looping,

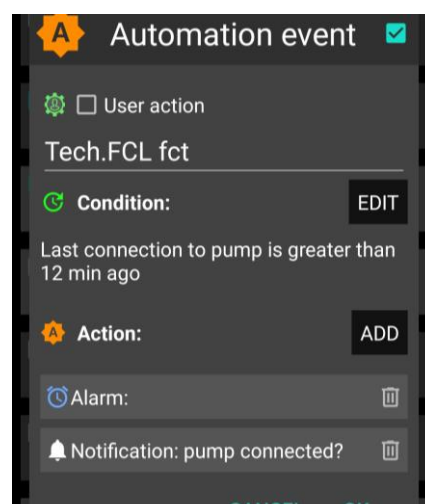
69

70 **Stable pump connection** (see [case studies 1.4](#) and [1.6](#))

71 In FCL you absolutely rely on your pump delivering,
72 without any further delay, the much needed insulin,
73 after any meal start.

74 Hence it is absolutely essential to avoid any
75 problems from a lost Bluetooth connection. In AAPS
preferences / Local alerts, switch alert on!

An Automation similar to the one pictured here →
could also help recognizing eventual problems.



1.4 Excellent CGM

You do not give a meal size-related bolus any longer. That leaves all insulination jobs to the algorithm! Around meals, a **stable Bluetooth** connectivity is absolutely essential, too, so CGM, loop, and pump can do their job without losing more valuable time (see [case study 1.4](#)).

As glucose values are the very basis for your autoISF loop, please **inform yourself well about your CGM:**

- How it principally performs (e.g. you absolutely must be “SMB always-ready” at cob=0)
- Whether you are using the best suited intermediate app that reports the “raw” value from the CGM transmitter into AAPS
- Specifically, how and where any smoothing is done, and what this might imply for the ISF boosting method you will be using See for instance here:
<https://androidaps.readthedocs.io/en/latest/Usage/Smoothing-Blood-Glucose-Data.html>
- Go through your data (in *all* day and also night *times*) to see whether your CGM produces any artefacts (jumpy values; see [case study 1.3](#)) that the loop could **misinterpret** as sign of a starting meal.
For some of these problems, e.g. “jumps” associated with nighttime compression lows, there are options to mitigate (see [section 5.1.2./3.\)](#) . See also the User Action Automation discussed about 2 pages below ([line 149ff](#)).
- In case your CGM requires calibrations: Note that calibrations often produce jumps. In that case, be prepared to do, in the future, an extra handling step to protect from your FCL reacting harshly.

autoISF has also a couple of in-built checks on the quality of the recent CGM values.

They are particularly important in the acceleration detection.

You will learn more about how a **“lousy” CGM performance can badly interfere with your tuning**, and resulting autoISF loop performance, in [section 4.2.4](#) (see red warning box there).

To summarize, a CGM with more scatter will make the loop lose more time, and lead to higher peaks and lower %TIR.

111 So, if you are unhappy with a slow reaction of your loop it could be because the loop is
112 unhappy with your CGM.

113 Consult the detail info given (at the time) in your SMB tab, or look it up later in the logfiles (using
114 the Emulator, [section 10](#), eventually).

115

116

117 1.4.1 Dexcom G6 and other 5-minute CGMs

118

119 The best proven way to stay out of trouble currently is to use Dexcom G5 or **G6**, and to ensure via
120 **overlapping** right and left arm sensor and transmitter utilization always good quality values, that
121 can be used by the Full Closed Loop ([case study 1.5](#)).

122

123 Other ways (using values from just *one* G6, or Dexcom ONE, G7, Libre2, or other new AAPS
124 integrated methods) are possible, but come with a lot of monitoring effort (best via watch), and
125 occasional time-outs for your FCL.

126

127 One safety feature in autoISF is a **blockage of SMB delivery whenever delta bg** (within the last
128 two 5 minute values) is **higher than 30% of that bg** (or higher +20%, at bg targets above 100
129 mg/dl).

130 Example: From 74 mg/dl, a jump to 97 (+23 mg&dl = + 31% of 74) or more would not
131 receive SMB "response". From 100 mg/dl to 131 mg/dl (+31) would neither.

132

133 Check in your (HCL or FCL) data whether at meals or sweet drinks with rapid absorbing carbs you
134 could run into the problem that jumps are "too high" and much needed insulin will be blocked (only
135 come via very much smaller portions.

136 For example, 400%TBR @ 0.6 U/h => 0.2 U in 5 minutes, instead of one ~3 U SMB. The
137 difference (of 2.8 U missed) translates @ ISF~ 40 mg/dl/U into up to + 112 mg/dl higher bg
138 peak! It will not become quite that bad, because the loop will catch up to the
139 insulinRequired with its next couple of decisions.

140

141 Instead searching in old data, you can also just have an eye on instances where you think a first
142 SMB was due, *but blocked*. Confirm that (by looking in the SMB tab) and think about a solution that
143 would not require changing the 30% safety limit in the code.

144 For instance, not drinking so much juice rapidly around meal start could be a "behavioral"
145 correction to get rid of the problem.

146

147 This *blockage* (no SMBs) would likely last only 5 minutes (and go probably unnoticed). However,
148 not only would you lose 5 valueable minutes to get your iob substantially elevated, but also, all

149 following deltas are likely much smaller. As a consequence, if the >30% delta was in fact (largely)
150 due to carb absorption, you would, just when needed most, miss some of the boost sought from
151 bgAccel_ISF.

152

153 This example also underscores that the CGM in use cannot be allowed random scatter that leaves
154 no reasonable room for safe detection of (smaller and) bigger “truly carb related” deltas

155

156 If or when (like: first half day of a new sensor) you are not sure about sufficient CGM
157 performance you might develop for yourself an Automation with User action ticked (along
158 the lines as used for other purposes in [section 5.2.2.3](#)). It would “ask you” before giving a
159 SMB whether you really want it delivered. That way you could a) have a look on your
160 glucose curve b) .. and on the ai % (underneath Autosens%), which indicates the relative
161 aggressiveness of ISF modulation from “what autoISF sees in your bg curve” c) think about
162 what sense a SMB now makes with respect to your past meal, and the carbs to be still
163 absorbed. Ultimately, you could also d) consult some of the detailed info given (every 5
164 minutes) in your SMB tab.

165 Such User action Automations need not be active at all times, but if you have it for
166 your first half day of a new G6 sensor for instance, you could activate that
167 Automation from your list of Automations; after the values have settled in, you can
168 de-activate (“shelve”) it again.

169

170 For a brief period, and if you are tech savvy, another way to deal with uncertainty about
171 CGM would be to employ the emulator method as presented in [section 11](#): Run a “too
172 mildly” tuned FCL, and in parallel run a “what-if” with your more aggressive settings that you
173 really would like to use once you are certain about your CGM.

174

175 However, I found it **easiest**, to **lay a solid groundwork by using** 1 Anubis, and **two**
176 **overlapping G6**, to get rid of most problems (...that I keep seeing in my data, on the worse
177 sensor of the two that often run for some days in parallel; see [case study 1.5](#)).

178

179 With a sensible **ioBTH** defined (see [section 2.4](#)), and your standard **alarms** for going towards a
180 hypo not silenced, the worst consequence from any automatically “over-treated” glucose jump
181 should be that you **need an unplanned snack** for the balance of “missing” carbs.

182 Like you should be used to from anti-hypo snacks also in your Hybrid Closed Looping past, you
183 might also in this case here prevent SMBs for a short while by setting a (here: odd numbered)
184 temporary glucose target (TT).

185

186 A disturbing and late sign of dealing with a too unreliable CGM (or too aggressive settings) could
187 be a trend towards increasing TDD and body weight during your early FCL experience!

188

189 1.4.2 CGM with 1 minute values (e.g. Libre3)

190

191 **Libre 3** showed promising performance too, and using it with **1-minute** values via Jugglucoo is
192 integrated, starting with autoISF version 3.0.1.

193

194 autoISF automatically detects whether values come through Jugglucoo, and adjusts parabola fit
195 calculations to determine acceleration etc. - It is currently too early to tell, but the 1 minute values
196 applied to the parabolic bg curve fit *could* present an avenue to earlier/better acceleration detection
197 than the 5-minute CGMs enable.

198

199 When running on Libre3 / 1 minute, you get many more SMBs that each are, on average, much
200 smaller. This has **implications on** some related **settings** (notably, the smb_range_extention, see
201 [section 2.1](#)).

202

203 Tests done prior to introducing the 1 minute Libre3 option (from autoISF 3.0.1 onwards) showed
204 overall comparable but smoother insulin delivery.

205 *Call for a [case study 1.7](#) from a Libre3 user!*

206

207 1.5 Meal-related limitations?

208

209 Setting up a full closed loop is relatively easy for people whose diet does not consist **mainly** of
210 components with rapid high effect on blood glucose (more see
211 <https://androidaps.readthedocs.io/en/latest/Usage/FullClosedLoop.html#meal-related-limitations>)

212

213 Meals do not have to be low on carb (provided you use a fast insulin for your FCL)
214 Fat or protein rich diets, or slow digestion/gastroparesis, make things easier rather than harder for
215 the full closed loop because late carbs nicely cover for inevitable “tails” of late action from SMBs
216 needed before or around peak time.

217

218 **Erratic consumption of smaller snacks with fast resorbing carbs can be a problem.**

219 In autoISF you can reduce this problem *to some extent* via one or two keystrokes from your
220 AAPS home screen. While certainly being a deviation from the FCL idea(I), this would be
221 one of the exceptional situations where you better do a quick “nudging” step from your “FCL
222 cockpit”. Details see in [section 5.2.1](#) and [case study 5.2](#)

223

224 **Really, there are no meal limitations.**

225 **The sketched more problematic options just force you to decide which extra “nudging”**
226 **efforts and/or worsened %TIR** (and/or “behavioral adjustments”, like less snacking) **you are**
227 **willing to accept.**

228 229 1.6 Lifestyle-related limitations?

230 231 Providing a technically stable system

232
233 Full closed looping requires a 24/7 technically stable system, especially regarding

- 234 • reliable CGM signals
- 235 • Bluetooth stability with the pump (see [case study 1.4](#) and [1.6](#))
- 236 • keeping your phone in sufficient proximity at all times
- 237 • avoiding (or at least early recognition of) occlusion (see [case study 1.1](#) and [1.6](#)).

238 This requires a habit (or, unlikely, permanent attention to details) like keeping all components well
239 charged and in close proximity; making cannula (or pod) changes always early enough to lower the
240 risk of occlusion; and having always potentially needed parts with you.

241 Depending on your system, your experience with it, but also on your acceptance and general
242 lifestyle, these aspects may or may not limit you.

243

244 Preparing for exercise

245

246 To prepare for exercise (sports, heavy work), the normal protocol with a pump or hybrid closed loop
247 is to take actions that reduce insulin on board prior to exercise

248 With your full closed loop, the algorithm is tuned to detect meals and to give you insulin to counter
249 glucose rises automatically. Setting a high temp. target and lower %profile, effective already
250 around meal start, could be a problem.

251 Unusual activity levels therefore likely require **disciplined preparation** (especially if you want to
252 **keep the need to snack during sports low**)

253

254 In autoISF you can reduce this problem to some extent via two or three keystrokes on your AAPS
255 home screen. While certainly being a deviation from the FCL idea(I), this would be one of the
256 exceptional situations where you better “flick a lever” from your “FCL cockpit” to keep iob low
257 (example see [case study 6.2](#)).

258

259

260

261 [Extra hurdles to establish FCL for kids](#)

262
263 To establish and maintain a FCL for kids brings about some extra challenges if:

- 264 • Lyumjev is not available or well tolerated
- 265 • Hourly basal rates are very low, providing a poor basis for big SMBs
- 266 • Diet is rich in sweet components. With the typical low blood volume of a small body, strong
267 tendency towards very high bg spikes!
- 268 • Going through marked changes of insulin sensitivity or of circadian pattern makes it difficult
269 to keep the FCL appropriately tuned.

270 This problem is about the same in FCL as it is also in Hybrid Closed Looping. How-
271 ever, now you might expect miracles from the FCL. This is not going to happen. You
272 still must be pro-active by setting suitable temporary % profiles. (These serve as a
273 basis, also for your autoISF FCL, which generally modulates the *at-the-time-valid*
274 profile_ISF, for further enhanced (or reduced) loop aggressiveness).

- 275 • Discipline may be poor regarding keeping 24/7 Bluetooth connectivity
- 276 • Discipline may be poor regarding keeping infusion sites perfectly running.
- 277 • Between kid and supervising parent it must be guaranteed, especially in the initial weeks,
278 that an eye is kept on whether the FCL is working about as expected.

279 More see [section 7](#).

281 282 [1.7 Time required for setting-up](#)

283
284 Lastly, before enjoying a functioning full closed loop, you need to have a period of a **some weeks**
285 with some free time and „free head“ for set-up –. Can you get, in the time you are willing to invest,
286 to a result that you consider good-enough is really the question. Depending on your „habits“, and
287 which – if any - compromises (like doing cannula/pod changes more often, never starting meals
288 when bg sits high and is not predicted to rush down soon ...) are you willing to make (and
289 everyday able to stick to), for the ease of not having to deal with assessing meals and bolusing for
290 them?

292 [1.7.1 Recommended structured, step-by-step approach](#) (see also beginning of [section 4](#))

293
294 Setting up your personal FCL using autoISF is a substantial project, for which you should **follow**
295 **the sequence as described** in this e-book.

296
297 But there is **no need to implement it fully in one step**.

- There is nothing wrong to go in your well running Hybrid Closed Loop mostly, while switching to FCL only for dinners, for instance, or only for weekend lunches, as a start.
- Once you found feasible settings, you can expand to other meal times...
- and lastly towards figuring out your best strategies for challenges aside from meal management, as we shall discuss in [sections 5.](#) and [6.](#)

There are **alternatives** to using autoISF **for FCL**, as well. See [sections 7](#) and [13.](#) for more info.

Notably FCL using AAPS Master and Automations (see in [section 13.1](#)).could be a much easier and more error-tolerant way of stepping into FCL. In a clinical study with 16 participants about 80% TIR was achieved without much tuning effort

Also, watch out for what a 2025 on-going FCL study with AAPS in Australia and NZ will show. See CLOSE IT trial protocol in: [Wilkinson T. et al, BMJ Open 2024;14:e078171. doi:10.1136/bmjopen-2023-078171](#)

To close the circle to where we had started; A very time consuming pre-requisite might actually be to **first sort out your Hybrid Closed Loop** ([section 1.1](#)), so your profile parameters are set „right“, and your “old” data really can serve as a blueprint for what, now, you would like *your loop* to do in FCL mode

If you feel you better do your homework first there, I highly recommend some of the material in the neighboring HCL repo: <https://github.com/bernie4375/HCL-Meal-Mgt.-ISF-and-IC-settings>

1.7.2 “Trial and error” fast track alternative

Note that if you had used dynamic parameters or special Automations („loops inside the loop“) this might have balanced some principal errors, but leaves you now *without a good starting point*, as now you must get rid of these over-patches (see also warnings at start of [section 4](#))..

Nevertheless, you will find FCL success stories also from loopers who continue(d) in that spirit, and just jump(ed) into using more powerful tools, in kind of a trial and error mode, frequently adding the latest add-on, or self-constructed patch (often in form of an Automation), to counter-balance encountered problems.

Resulting solutions may be good-enough.

However, they tend to be unstable and not well-understood. That is a poor basis for managing arising problems, and for temporarily adjusting to special situations.

335 Nevertheless, this is an alternative avenue for the impatient, less analytically, and more
336 adventurous inclined.
337 Note though, that it is hard to consult (help) someone who, over time, constructed his/her
338 own complicated maze of constructs.
339 Still, I welcome everyone's creativity. "Not macht erfinderisch" (necessity is the mother of invention)
340 we say in German: Something amazing of broader interest could emerge, and help push
341 innovation for us all.

342

343 Be prepared to needing **a fresh re-start, in case your trial-and-error got you completely lost.**
344 Depending on knowledge level and experience, this could happen on a "fast track" route.

345

346 1.7.3 Safety first

347

348 Regardless which route you choose, PLEASE always observe the **safety** settings/instructions
349 coming with the DIY dev- variant of FCL software you select.

350

351 All FCL methods come with boosted SMBs. So a key safety measure every user going towards
352 any FCL should have in place is to set an **iob threshold** (iobTH; size a bit below what you used as
353 a bolus for bigger meals in HCL) above which no more SMBs can be given by your FCL.

354

- 355 • iobTH is an integrated feature of *autoISF* (see [section 2.4](#)).
- 356 • *Other AAPS-based FCL methods* may require to set up an Automation for a temporary iob
357 threshold that blocks SMBs from being delivered, see e.g. here for AAPS FCL
358 w/Automations; [https://androidaps.readthedocs.io/de/latest/Usage/FullClosedLoop.html#iob](https://androidaps.readthedocs.io/de/latest/Usage/FullClosedLoop.html#iob-threshold)
359 [b-threshold](https://androidaps.readthedocs.io/de/latest/Usage/FullClosedLoop.html#iob-threshold) .
- 360 • *In case there are other methods than autoISF* for FCL also on the *iAPS* or *Open iAPS* platforms, you
361 may have to rely on an adjusted iob_max border, or watch the iob development, and intervene with a
362 SMB shut-off, or by opening the loop, when deemed necessary.

363

364 Also, make use of the easy-to-use feature of **SMB shut-off at odd** profile or temporary **target**. This
365 can **at any time easily** be done manually, via the top right "TT" field in your AAPS screen (set, and
366 time, an odd-numbered target; [section 5.1.3](#)), and can be of enormous help **to temporarily**
367 **safeguard you from aggressive loop actions** (i.e. further growth of iob, no matter how close you
368 already may be to the iobTH)-

369 This is the same concept that you already know from your HCL times, when you wanted to "tame your loop"
370 so it does not "fight" your anti-Hypo Snack with a SMB (An elevated TT> 100 mg/dl was then used, to shut
371 off SMBs in HCL for a while).

372 Lastly, make sure you “train”, for your set-up weeks, how/when to **transition between FCL and**
373 **your prior HCL** mode (refer to [section 5.2.3](#) and [5.1.1](#)).
374 A new User Interface has been suggested to ease this transition via a modified loop button in the AAPS main
375 screen (developers: see [section 5.4,1](#)).